WASP-36b

R-1760

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-36_190414` · created 2026-08-02T20:53:52+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458587.6918 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.047
Transit depth (Rp/Rs)^2	2.62 +/- 1.54 [%]
Orbital Inclination (inc)	83.83 +/- 2.91
Airmass coefficient 1 (a1)	1.39 +/- 0.7
Airmass coefficient 2 (a2)	-0.18 +/- 0.28
Scatter in the residuals of the lightcurve fit is	51.73 %
Best Comparison Star	None
Optimal Aperture	1.93
Optimal Annulus	3.0
Transit Duration (day)	0.077 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.047 vs 0.1368 (deviation 0.54σ)
Duration fit vs published	0.077 d vs 0.0758 d (deviation 0.05σ)
Mid-time O–C	7.5 min
Depth SNR	3.4
Residual scatter	51.73 %

Light curve

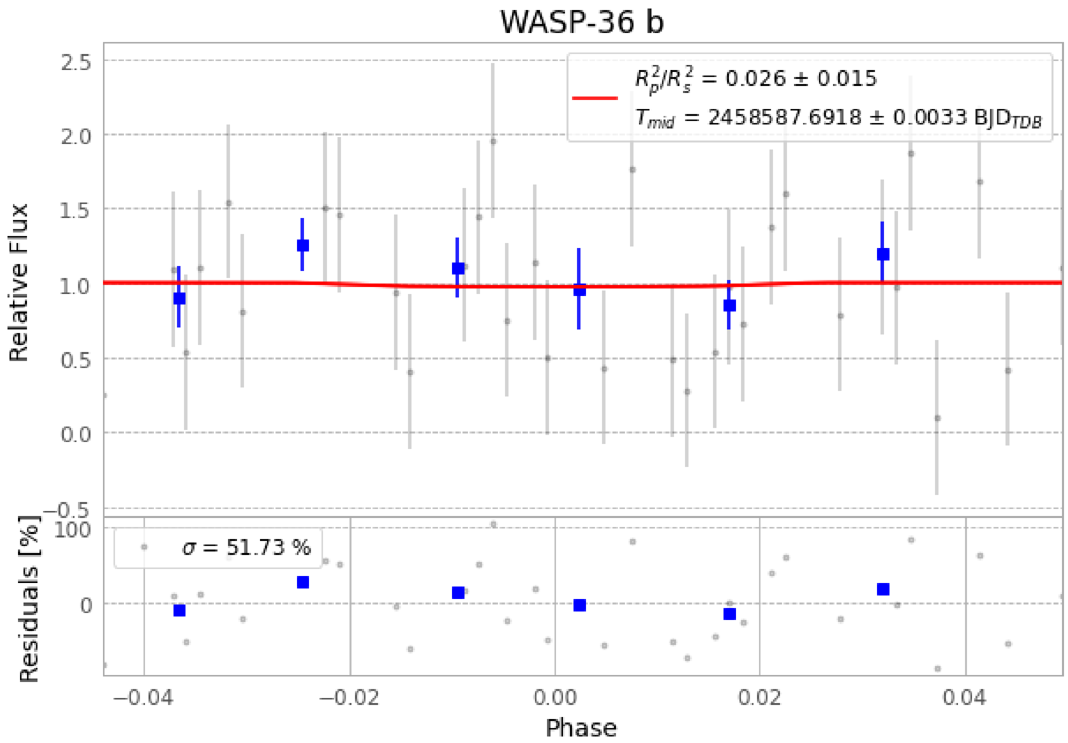

Figure 1 — FinalLightCurve_WASP-36 b_2019-04-13.png (SHA-256 3cd1d346bc76c549…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [521, 360], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1271cb493198e121…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

68 FITS frames (45 MB), `WASP-361904140254O9.FITS` → `WASP-361904140621I2.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-190414.log (2d515efb461d…), FinalLightCurve_WASP-36 b_2019-04-13.png (3cd1d346bc76…), FinalLightCurve_WASP-36 b_2019-04-13.csv (750b67475aaf…)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-08-02 20:53Z.